BALOGUN MIRACLE ABIODUN

Instagram Login Page – Social Media Phishing Simulation

Objective

To simulate a social media phishing attack targeting an Instagram login page, in order to understand how such attacks are carried out and to document how employees can recognize and defend against them.

Target: Simulated Instagram login page (test environment only — no real accounts or real victims were targeted)

Tool: Zphisher v2.3.5 (Kali Linux)

Tunnelling Service: Cloudflared (auto-generated HTTPS tunnel)

Procedure

A controlled phishing simulation was carried out using Zphisher on Kali Linux to clone the Instagram login page and host it behind a temporary Cloudflare tunnel. A custom masked URL was generated and opened in a browser, where made-up test credentials were submitted. Zphisher's terminal output was monitored to confirm that the simulated "victim" visit and submitted data were captured, demonstrating how a real credential-harvesting phishing page operates end-to-end.

Zphisher Setup

Zphisher was cloned from its official GitHub repository, made executable, and launched from the terminal on Kali Linux.



Figure 1: Cloning the Zphisher repository and preparing the script (chmod +x zphisher.sh)



Figure 2: Launching Zphisher with bash zphisher.sh (packages checked / Cloudflared installation)

Selecting the Social Media Template (Instagram)

On launch, Zphisher presented its attack-template menu covering a range of popular platforms. Option [02] Instagram was selected, followed by the "Traditional Login Page" template.

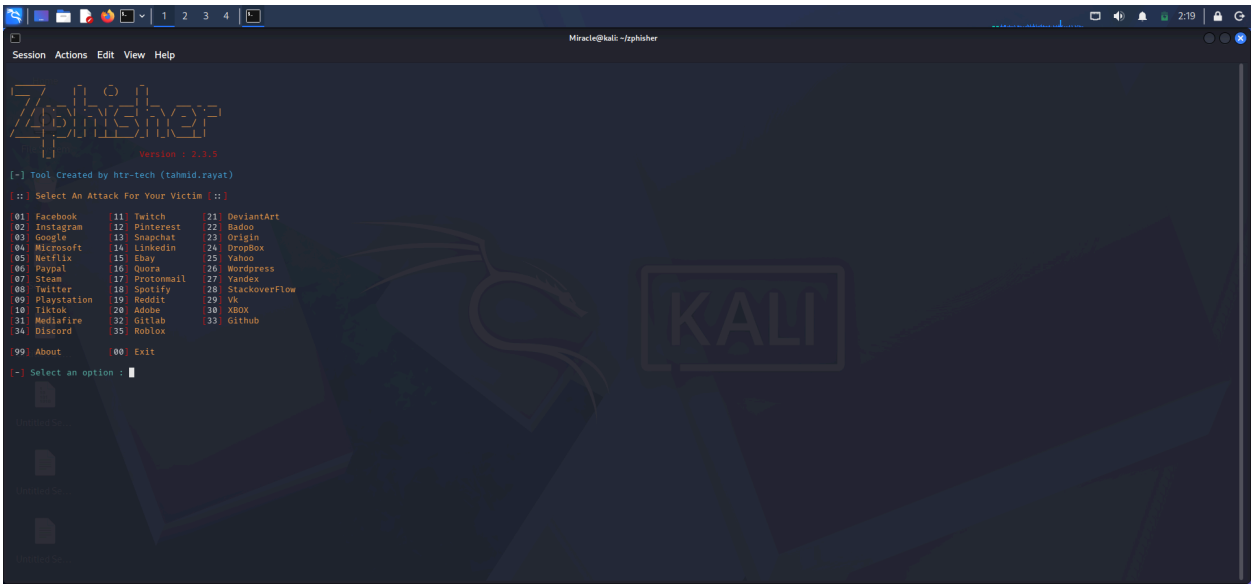


Figure 3: Zphisher main menu — Instagram (option 2) selected as the target platform

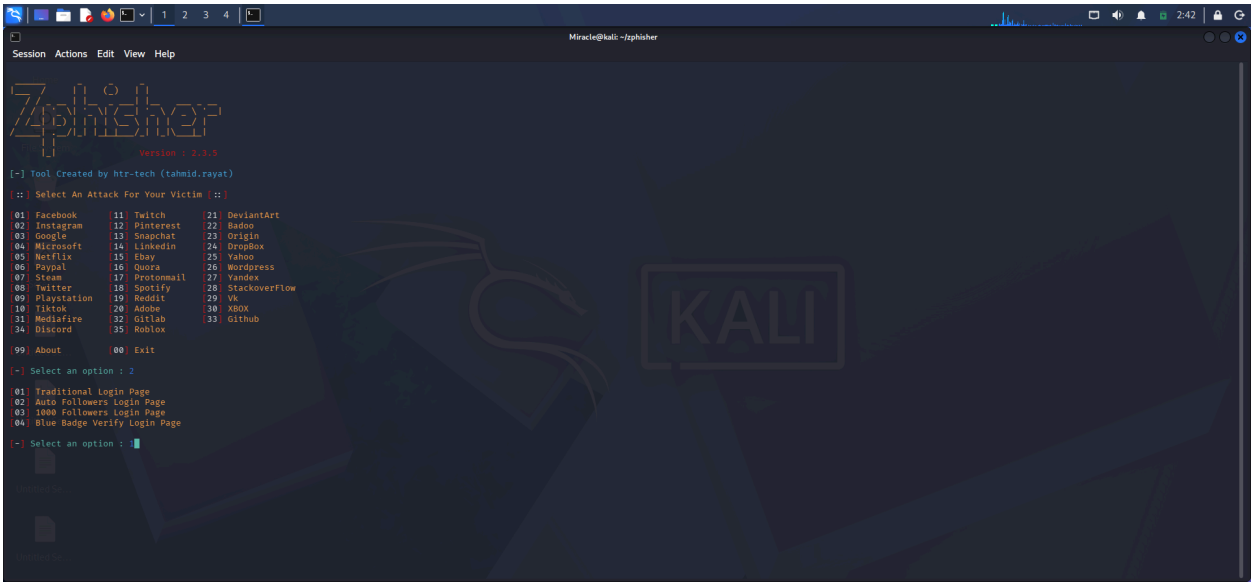


Figure 4: Instagram login-page template options — "Traditional Login Page" selected

Generating the Phishing Link

Cloudflared was selected as the port-forwarding/tunnelling service so Zphisher could auto-detect a public HTTPS link. A custom masked URL was then created to make the generated link appear more convincing, as an attacker would.

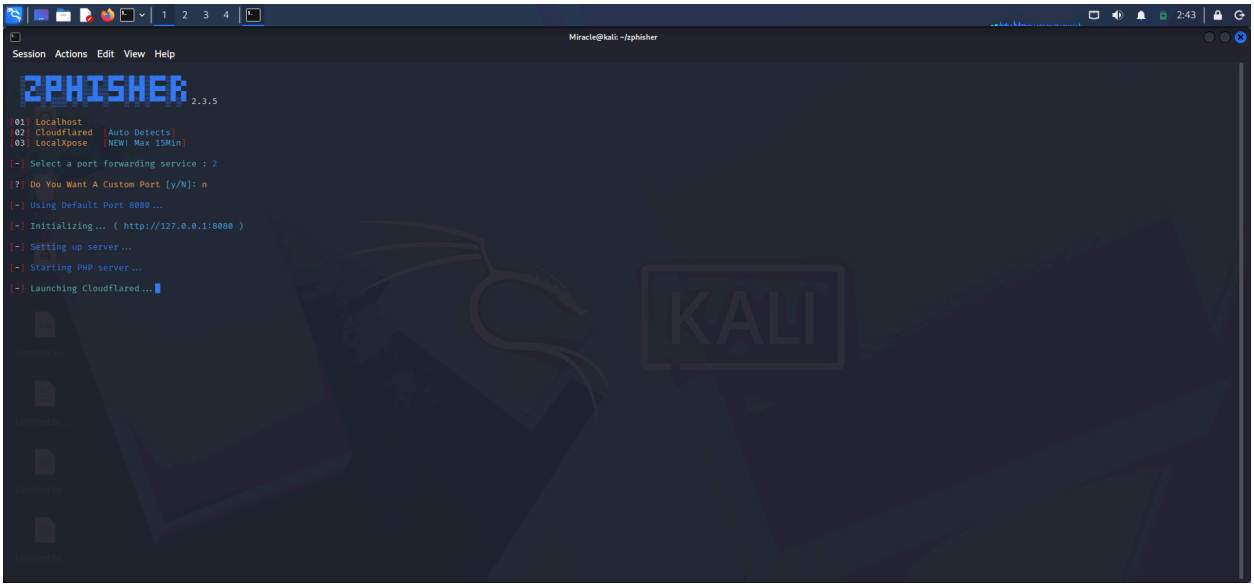


Figure 5: Port-forwarding menu — Cloudflared selected to generate a public tunnel

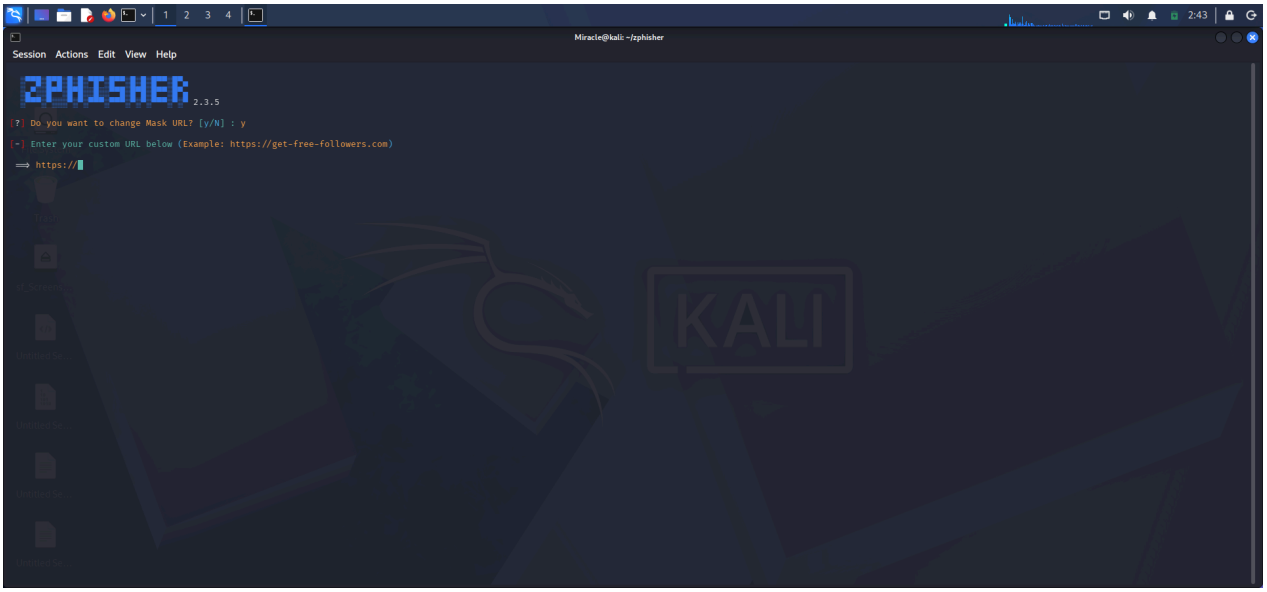


Figure 6: Creating a custom masked URL for the generated phishing link

. Phishing URL Generated

Zphisher produced the live phishing link and began listening for incoming visits. Once the link was opened, Zphisher's terminal confirmed the connection, logged the visiting IP address, and captured the submitted credentials.

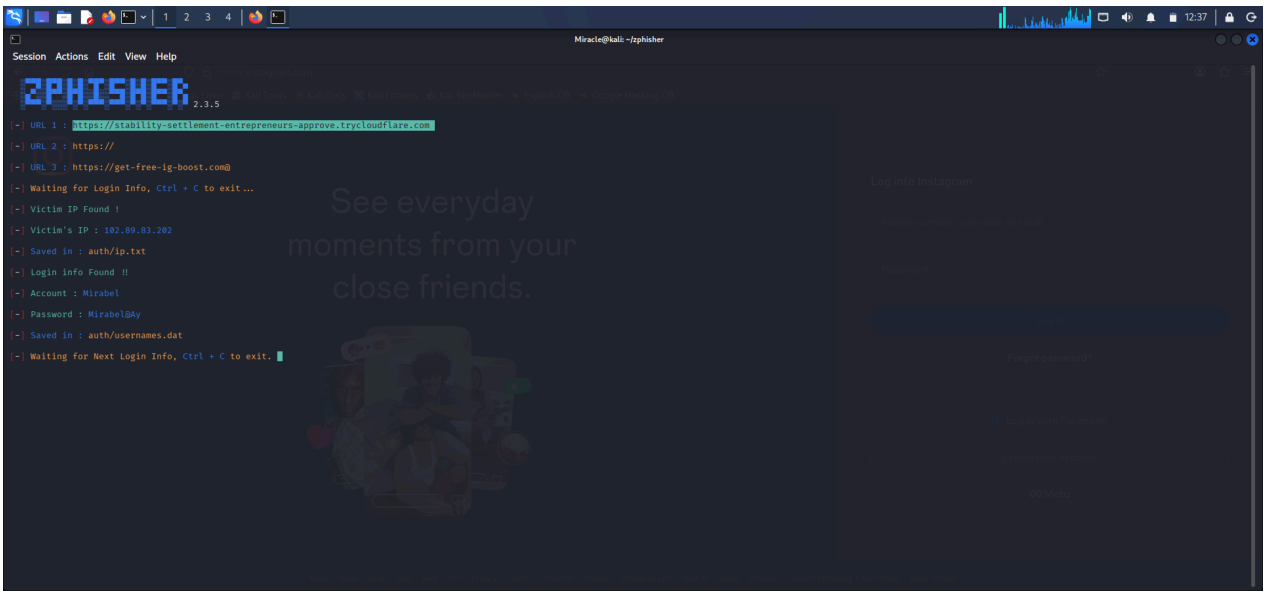


Figure 7: Generated phishing URL (Cloudflare tunnel), victim IP captured (102.89.83.202), and test credentials logged

Fake Login Page Rendered in Browser

The generated link was opened in a browser, rendering a page that is visually identical to the real Instagram login screen — confirming that an unsuspecting employee would have no visual indication the page is fraudulent.

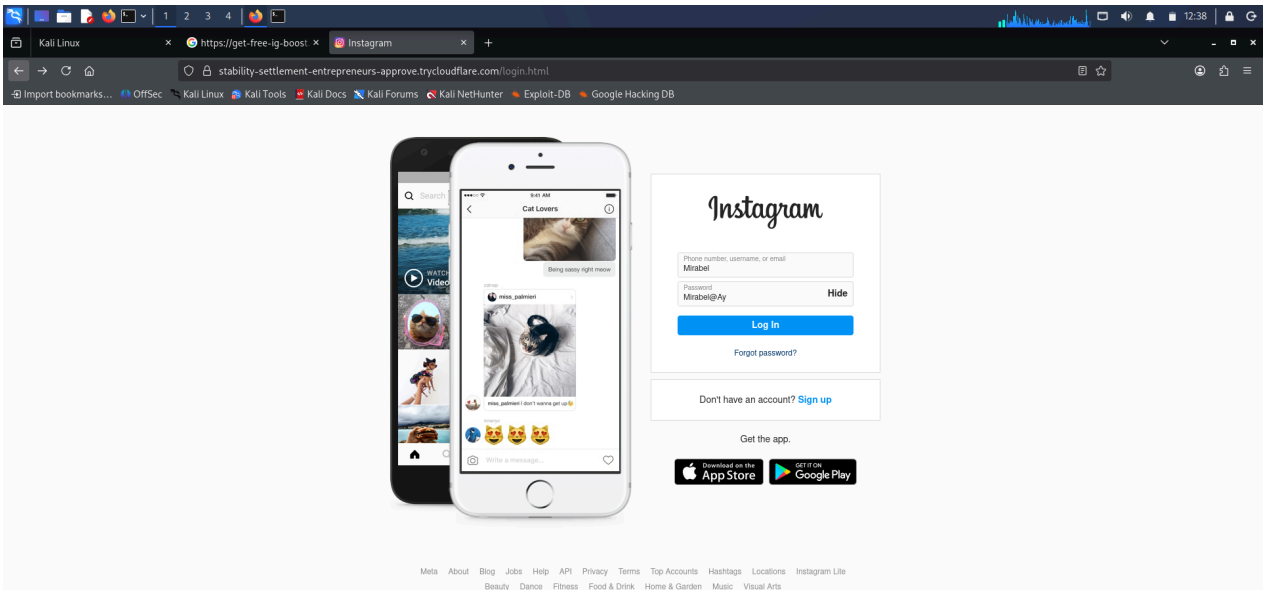


Figure 8: Cloned Instagram login page rendered in the browser via the generated phishing link

. Capturing Test Credentials

Made-up test credentials (username “Mirabel”, password “Mirabel@Ay” — not a real account) were entered into the cloned form to simulate a victim submitting their login details. The credentials and the visitor’s IP address were successfully captured by Zphisher and written to local files (auth/ip.txt and auth/usernames.dat).

4. Analysis of Results

The simulation confirms that a convincing, pixel-for-pixel clone of a social media login page can be produced and distributed within minutes using freely available tools, and that it is capable of transparently logging visitor metadata (IP address) and any credentials typed into it. Because the page is a static copy of the real site, there is no technical flaw being exploited on Instagram’s side — the attack succeeds purely through deception of the end user. The only reliable tells available to a victim are the URL/domain shown in the address bar (here a trycloudflare.com tunnel domain and a masked “get-free-ig-boost” style link), the absence of HTTPS trust indicators for the real brand domain, and browser-level reputation warnings.

5. What is Social Media Phishing?

Social media phishing is a social-engineering attack in which an attacker creates a fake but convincing copy of a social media platform’s login page (e.g., Instagram, Facebook, Twitter) and tricks a target into visiting it — typically via a disguised or “masked” link sent through email, SMS, or direct message. When

the victim enters their username and password, believing they are logging into the real service, the credentials are captured by the attacker instead, granting them access to the victim's real account.

6. How Employees Can Identify and Avoid Social Media Phishing Attacks

• **Check the URL carefully before entering any credentials** — a cloned page will be hosted on a domain that does not match the real platform (e.g., a Cloudflare tunnel or masked link rather than instagram.com), even if the page looks identical.

• **Heed browser security warnings** — modern browsers flag known or suspicious phishing domains as "Dangerous" or "Not secure"; never proceed past such a warning.

• **Never click login links from unsolicited messages** — type the platform's address directly into the browser or use the official app instead of following a link shared by email, SMS, or DM.

• **Enable multi-factor authentication (MFA)** on all social media accounts, so a captured password alone is not enough for an attacker to gain access.

• **Use a password manager** — these only auto-fill credentials on the legitimate, saved domain and will not populate fields on a look-alike phishing site.

• **Report and verify suspicious links** through official IT/security channels before interacting with them, and verify unexpected requests for login through a separate, trusted channel.

7. Conclusion

This controlled exercise successfully demonstrated how Zphisher can be used to clone an Instagram login page, host it behind a disguised public link, and capture test login data end-to-end — purely for educational, defensive-awareness purposes within an isolated lab environment. The exercise reinforces that social media phishing relies entirely on deceiving the user rather than exploiting a technical flaw, and that URL verification, browser warnings, and MFA remain the most effective defenses available to employees.